



Think College Research Summit

December 7, 2017



Today's agenda

- Introduction to today's Research Summit
- Augmented Reality for Teaching Science Vocabulary to Postsecondary Education Students with Intellectual Disability and Autism -Don McMahon
- Strategy Instruction for Young Adults with Intellectual Disability at the Postsecondary Education Setting – Youjia Hua and Suzanne Woods-Groves
- Discussion



The TPSID National Coordinating Center at Think College is a project of the Institute for Community Inclusion at UMass Boston.
The Center is funded by the Office of Postsecondary Education Grant #P407B100002.





Theoretical Foundations of Augmented Reality

What is Augmented Reality?

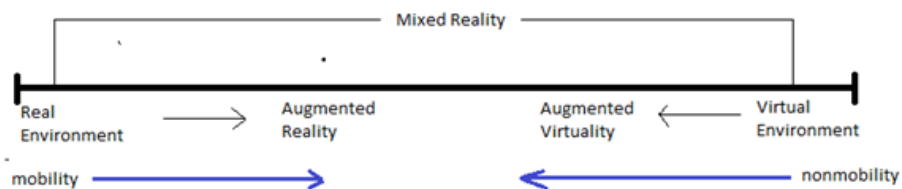
AR in Relation To UDL, AT, and IT

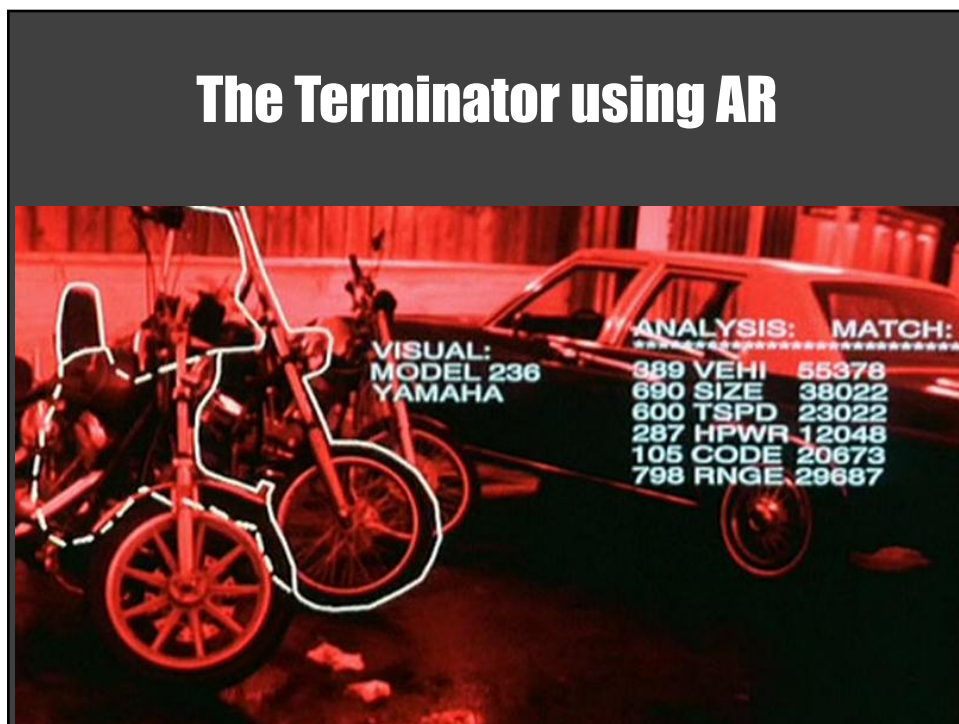
What is Augmented Reality?

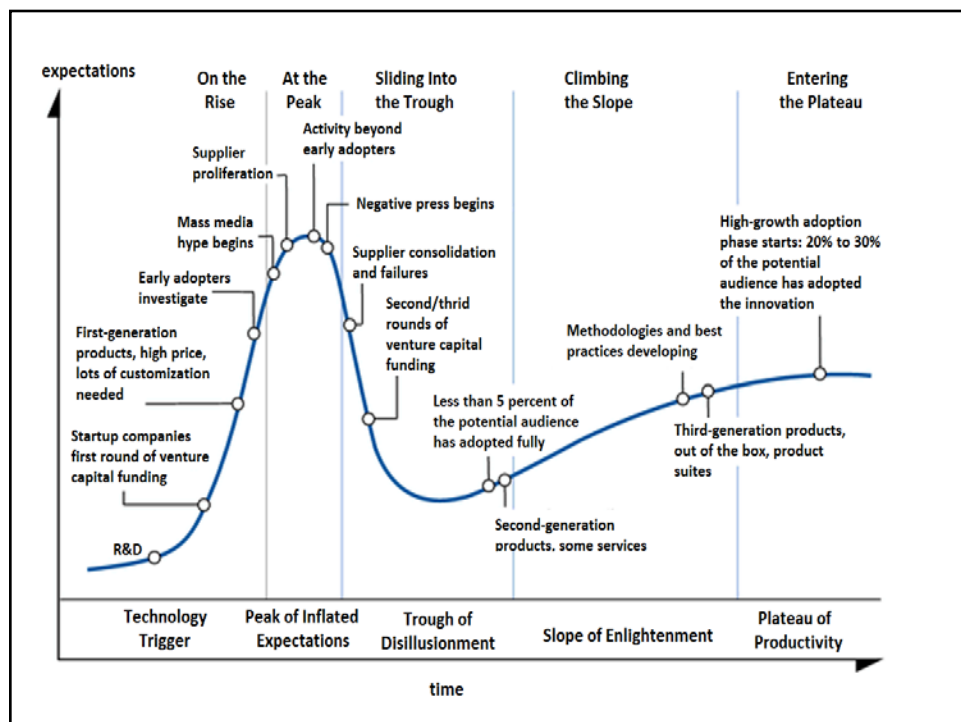
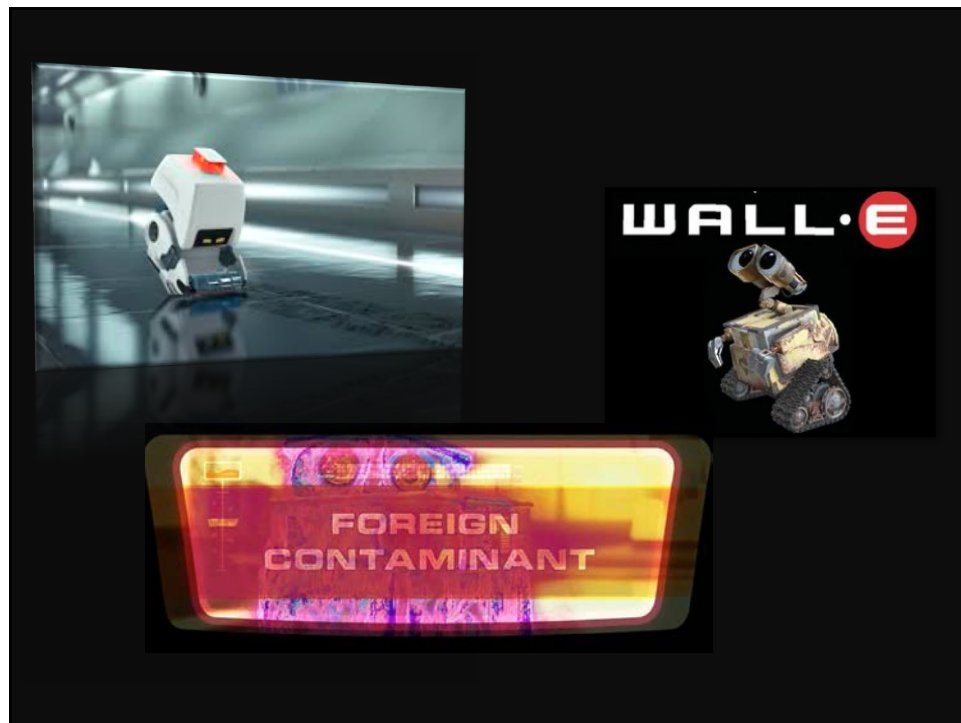
Milgram and Kishino (1994) in their taxonomy of mixed reality described AR as any instance where the “display of an otherwise real environment is augmented by means of virtual (computer graphic) objects” (p. 2).

AR is a medium using technology to bridge the digital world and physical world to create a blended experience that has the benefits of both (Craig, 2013).

Mediated Reality Continuum







Augmented Reality as an Instructional Tool for Teaching Vocabulary to Postsecondary Education Students with Intellectual Disabilities and Autism

- Examines AR as a tool for vocabulary instruction in science.
- Uses a multiple probe across skills design



Brief Review of Literature

According to the National Longitudinal Transition Study 2, no students with intellectual disabilities reported working in computers, engineering, or science related jobs. Whereas 3.8% students with other disabilities found employment in these fields (Newman, et al., 2011).

Reading instruction for students with ID and ASD has traditionally overemphasized functional reading skills such as sight words necessary for daily living, safety, and independence (Browder, Wakeman, Spooner, Ahlgrim-DeLzell, & Algozzine, 2006; Kliever, 1998).

Literature Review continued

A fundamental aspect of developing literacy skills is acquiring vocabulary. Bell and McCallum (2008) established a strong connection between vocabulary proficiency and increased reading comprehension, writing skills, and overall academic achievement of students with disabilities.

Research Questions

The purpose of this study is to examine the effects of a marker-based AR technology to teach college students with ID and ASD science related vocabulary words. Specific research questions include:

- What are the effects of marker-based augmented reality vocabulary instruction on the acquisition of science vocabulary words of college students with ID and ASD?
- Do college students with ID and ASD find augmented reality vocabulary instruction to learn new science vocabulary words socially acceptable?

Methods Study 1

A multiple probe across behaviors/skills design (Gast, 2010) was used to examine the relation between the AR based vocabulary intervention and each participant's performance to correctly identify and label the meaning of the science vocabulary word.

Participants

- Four individuals with an intellectual disability or autism attending a postsecondary education program participated in this study. Participants ranged in age from 19 to 23 years old.
- Participants were selected based on the following (a) diagnosis of an intellectual disability or autism (b) participation in a postsecondary education program, (c) no physical disability that impeded the performance of the skill, and (d) consent to participate in the study.
- Participants' full-scale IQ standard scores ranged from 48 to 85. All students received special education services under the ID category during their K-12 schooling.

Participant Descriptions Study 1

Participant descriptions Study 1.

Participant	Age	IQ	Woodcock Johnson III (Standard Scores/Proficiency)			Brigance Transition Inventory (Grade Equivalents)	
			Processing Speed	Basic Reading	Reading Comp.	Decoding	Vocab Comp.
Miguel	25	85 ^A	81 / Limited	100 / Average	82 / v.limited	8 th	4 th
Catherine	25	48 ^B	50 / v.limited	41 / negligible	55/ negligible	4 th	2 nd
Billie	19	67 ^C	68 / limited	73 / v.limited	71/ v.limited	6 th	2 nd
Brenda	20	61 ^D	52 / v.limited	59/ v.limited	70/ v.limited	5 th	3 rd

A= Wechsler Adult Intelligence Scale III (WAIS III), B= Wechsler Intelligence Scale for Children (WISC III), C= Reynolds Intellectual Assessment Scales (RAIS), D= Kaufman Brief Intelligence Test 2 (KBIT2).

Setting

- Participants attended a postsecondary education program (PSE) for individuals with intellectual disabilities and ASD located at a national university.
- Each participant was enrolled in traditional university courses for audit credit, recreational classes, student work internship, and program specialty/core courses that included life skills, career development, and digital literacy.
- Each participant was included in traditional university courses and activities for a minimum of 80% of the week. The core courses were designed specifically for college students with intellectual disabilities enrolled in the PSE program.
- All phases of this study occurred in a computer lab located on campus.

Materials: Vocabulary Assessment

Vocabulary tests were developed to assess the three science-related word lists: (a) human bones, (b) human organs, and (c) cell biology. Ten target vocabulary words were identified for each word list by the investigator based off low pretest knowledge by the participants. Each vocabulary test was a 20-item assessment that included two questions for each of the 10 vocabulary words on the list. This assessment was intended to measure understanding of the vocabulary terms by measuring both the ability to define and correctly label the selected science terms.

Definition Questions = 10 Multiple Choice

- Measured the ability to match a description/definition to the correct vocabulary term
- Definitions were adapted to simplify language
- Multiple-choice format with definition provided and four vocabulary word options
- The three distractor answers were all from the word list being assessed

Labeling Questions = 10 Label A Picture with the Correct Vocabulary Term

- Required the student to label either a diagram or figure with the vocabulary term
- A word bank of targeted vocabulary words was included
- Pictures used were royalty free images selected by the investigator and modified if necessary (i.e. arrows pointing to a specific structure)

Materials: Aurasma App

The app uses live video from the mobile device's camera to identify an object, in this case a printed marker.

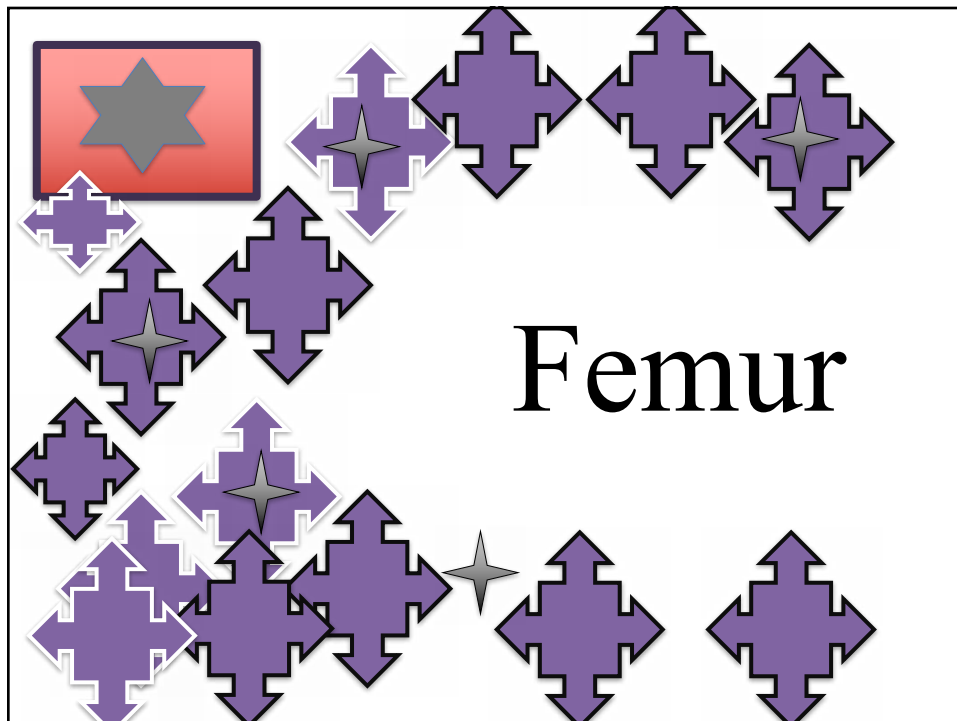
This augmented reality view then provides the user a view of the vocabulary word card and digital information in the form of pictures and video that identify the meaning of the word.

During intervention phase, an iPad equipped with the Aurasma app was provided to each participant.

Materials: AR Triggers

The investigator created one trigger image for each of the 30 science vocabulary terms in the study.

- Each trigger included a large print (72 point font) of the vocabulary term and a unique design comprised of different shapes in order to provide enough detail for the app to “recognize” it.
- Triggers were produced using Microsoft’s PowerPoint and then printed as handouts.
- The printed triggers were stapled together to create a 10-page book of “AR vocabulary cards” for each word list.



Materials: AR Content

The investigator created one trigger image for each of the 30 science vocabulary terms in the study.

The AR content displayed was a short 30 to 35 second video created by the investigator for each vocabulary term. The elements of each video included:

1. Title slide of the vocabulary term
2. Video with audio of the definition text being read aloud electronically
3. The same free to use image used in labeling activity for the vocabulary term with the correct vocabulary term labeled
4. Video of a 3D simulation showing the location of the vocabulary term, during which the audio from the definition being read aloud was repeated
5. Repeat of the image of the vocabulary term as shown in the labeling part of the assessment with the audio of the definition being played a third time

Videos either were taken by the investigator (definitions, bones, and human organs) or were used with permission (i.e., parts of the plant cell).

Variables and Data Collection

- The **independent variable** was the use of the augmented reality app to learn new vocabulary.
- The **dependent variable** was defined as the percentage of correct responses on the vocabulary assessment for each of the three word lists.
 - Each session was made up of an assessment that included 20 items for each of the three science word lists.
 - The vocabulary questions were ordered randomly on the assessments to reduce practice effects.
 - If the participant correctly answered the question, then it was recorded as a correct response.
 - If the participant did not answer the question correctly or did not respond, then it was recorded as an incorrect.
 - The number of correct responses was divided by the total number of items available to calculate the percentage of correct responses per session, which was then graphed for each participant for visual analysis.

Procedures

Baseline.

- During baseline, each participant completed 3 sets (human bones, human organs, cell biology) of the 20-item science vocabulary test targeting 10 vocabulary words.
- Although the test was read aloud, no additional feedback or prompts were provided. Participants were instructed to answer the vocabulary questions on the assessment and were told they could skip questions they did not know.
- Stability was determined using the “80%-20%” criteria of stability envelope (Gast, 2010). If 80% of the data points fell on or within 20% of the mean of baseline, the data would be considered stable.

AR Vocabulary Intervention.

- At the start of each intervention session, students completed the vocabulary assessment.
- Afterwards, they were given three opportunities to use AR to practice learning the science vocabulary words. Students were given the vocabulary words, iOS device, and instructed to “try and beat the definition”.
- That is, students would try to verbally define the word before the AR definition was displayed. Students also were instructed to repeat the AR definition after viewing the AR definition display. After the students practiced the first vocabulary word, they proceeded to the second vocabulary word and so forth until all ten words were practiced.
- Students then practiced all ten words two additional times for a total of three practice opportunities. The students continued to practice defining the anatomy vocabulary words until they performed 100% on three consecutive quizzes.

Data Analysis Procedures

- Visual analysis procedures were used to evaluate the results of the multiple baseline across skills intervention using augmented reality to teach science vocabulary. Intervention effects were assessed using six indicators to examine within-and between-phase data patterns:
 - (a) Level
 - (b) Trend
 - (c) Variability
 - (d) Immediacy of the effect
 - (e) Overlap
 - (f) Consistency of data patterns across similar phases (Kratochwill, et al., 2010).

First Effect Size -PND

- For each participant the percentage of non-overlapping data (PND) was calculated between the baseline and intervention phases (Scruggs, Mastropieri & Casto, 1987).
- Based off those guidelines from this study evaluated PND greater than 70% as a highly effective intervention, PND greater than 50% and less than 70% as questionable effective, and PND less than 50% was considered unreliable effectiveness for interventions.

Second Effect Size –Modified Cohen's d (d_1)

The second effect size measure calculated was a variation of Cohen's d (1988) as calculated by Busk and Serlin (1992, p. 197).

Cohen's original effect size interpretational benchmarks were greater than .8 was a large effect size, greater than .5 and less than .8 was medium effect size, and greater than .2 and less than .5 was a small effect size.

Robey et al. (1999) revised these benchmarks for single-subject research and suggested 2.6 to 3.9 for small effect size, 3.9 to 5.8 for medium effect size, and greater than 5.8 as a large effect size.

Social Validity

- Following the conclusion of the intervention phase, each participant was asked to complete a Likert survey (See Appendix) created by the investigator to assess their opinions and acceptability of using the AR intervention to learn new vocabulary.
- The question items were read aloud to the students. Each survey item used Likert scale ranged from 1 (strongly disagree) to 5 (strongly agree) with the addition of "Sad Faces" (1 Strongly Disagree) to "Smiling Faces" as indicators on the scale to support comprehension of the question. The social validity survey also included two open ended social validity questions whose answers were scribed by the investigator.

MIGUEL

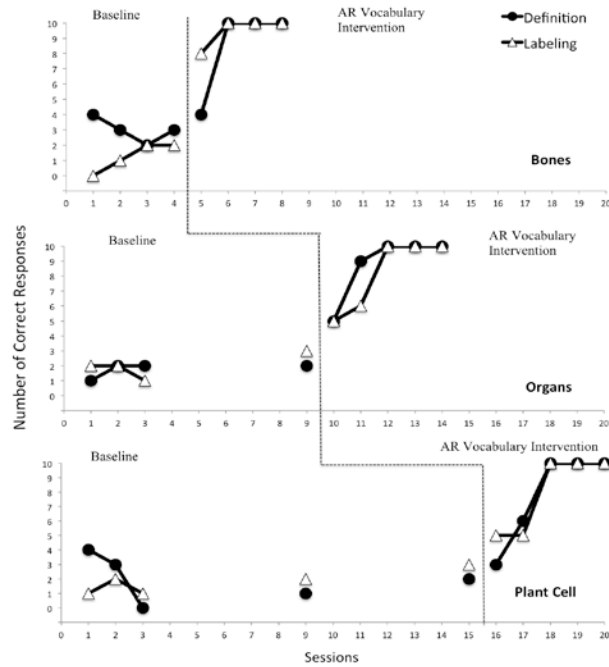
Across all conditions, Miguel immediately improved his science using the AR vocabulary intervention.

During the AR intervention, on the first word list, Miguel reached criteria of 80% correct definition and labeling responses for three consecutive sessions on his fourth session.

On the second word list, he reached criteria after his fifth session using the AR vocabulary instruction.

On the final word list, he reached criteria after his fifth session using the AR vocabulary instruction.

Visual analysis shows that his definition score and labeling score improved at approximately the same rate.



Catherine

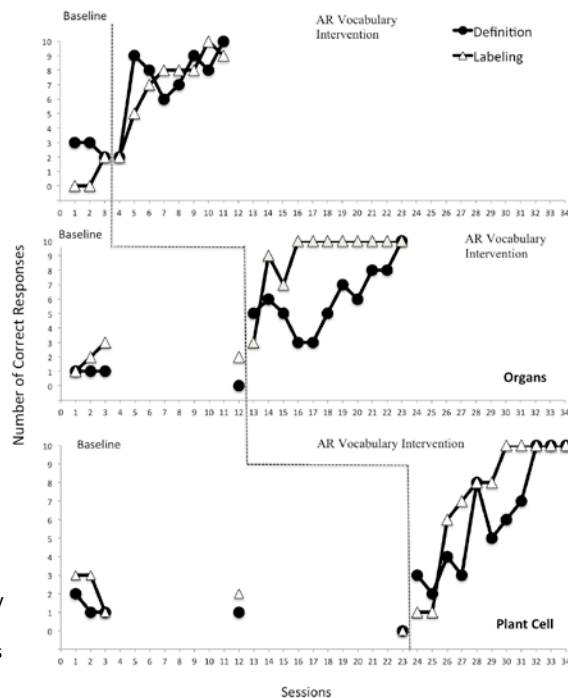
Catherine learned the three sets of science vocabulary terms using the AR vocabulary instruction.

On the first word list AR intervention, Catherine reached criteria of 80% correct definition and labeling responses for three consecutive sessions after her eighth session.

On the organs word list, she reached criteria after her eleventh session using the AR vocabulary instruction.

On the final word list, using the AR vocabulary experience she reached criteria after her eleventh session using the AR vocabulary instruction.

Visual analysis shows that her definition score and labeling score improved at approximately the same rate for the bones word list but her ability to correctly label improved faster than her ability to correctly find the definition on the organs and parts of the plant cell.



Billie

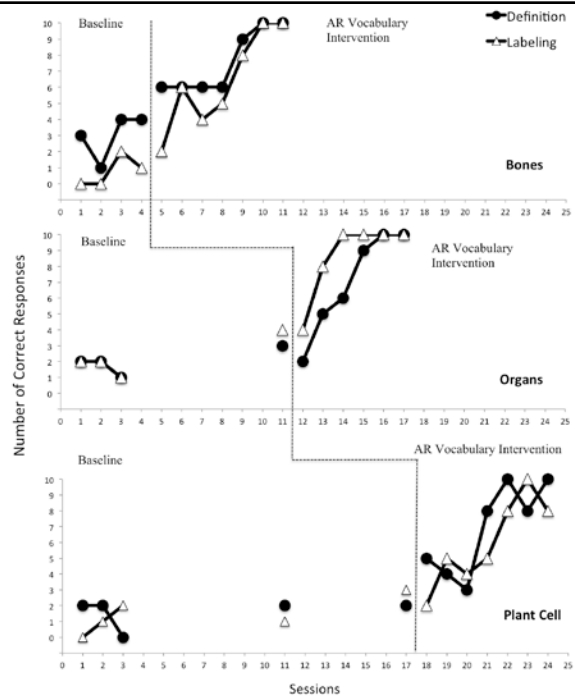
Billie learned the three sets of science vocabulary terms using the AR vocabulary instruction.

During the AR intervention, on the first word list Billie reached criteria of 80% correct definition and labeling responses for three consecutive sessions after her seventh session on the bones word list.

On the second word list she reached criteria after her eleventh session using the AR vocabulary instruction.

On the final word list, using the AR vocabulary experience she reached criteria after her seventh session using the AR vocabulary instruction.

Visual analysis shows that her definition score and labeling score improved at approximately the same rate for all of the word lists.



Brenda

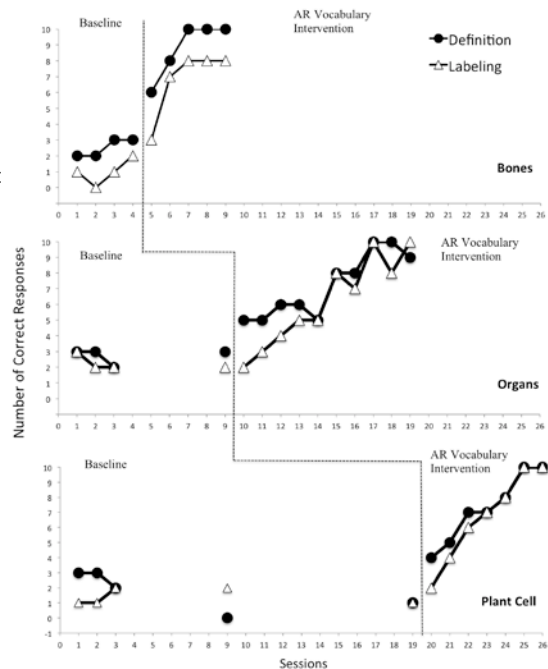
Brenda learned the three sets of science vocabulary terms using the AR vocabulary instruction.

During the AR intervention, on the first word list, Brenda reached criteria of 80% correct definition and labeling responses for three consecutive sessions after her seventh session.

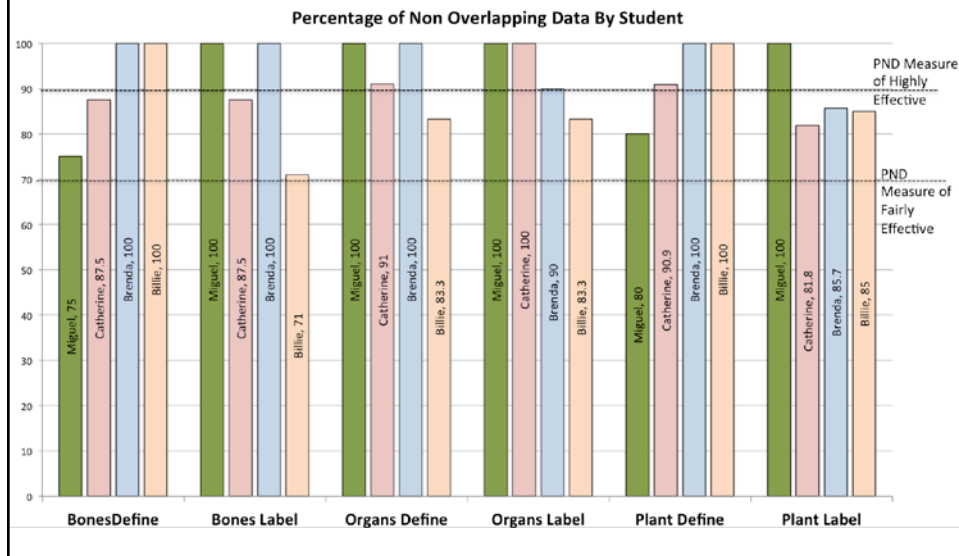
On the second word list, she reached criteria after her eleventh session using the AR vocabulary instruction.

On the final word list, using the AR vocabulary experience she reached criteria after her seventh session using the AR vocabulary instruction.

Visual analysis shows that her definition score and labeling score improved at approximately the same rate.



Effect Size Measure Percentage of Non Overlapping Data (PND)



Student	Questions	
	Open ended Question 1. What was it like to use the augmented reality vocabulary instruction?	Open ended Question 2. What did you like or not like about the augmented reality vocabulary instruction?
Catherine	"You point the camera at the paper and it explains it"	"This really is helping me learn my science"
Brenda	"The app made the words come to life and showed what they did"	"Seeing the video and the word together helped me to learn where they go [to be labeled]"
Miguel	"The definition just popped up from nowhere with pictures telling about the word."	"How it just popped up out of nowhere. How did it do that?"- Miguel
Billie	"It helps you learn the science [vocabulary words]"	"I liked how the pictures float above the science word. So you can learn about it."

Discussion: AR Vocabulary Instruction

- AR vocabulary instruction for students with disabilities works 😊
- These findings support previous findings demonstrating AR as an effective tool for science instruction.

Limitations

- One of the limitations of this study, like all single subject research is the small sample size of this study ($n = 4$) limits external validity and generalizability.
- All of the participating students attended a PSE program. Students were highly motivated adults with disabilities. Students had relatively strong basic computer skills. All students were familiar using mobile devices.
- AR was new to the students, but they often used computers and mobile devices for learning. The assessment of the science vocabulary terms included the students identifying the correct term from a list of four choices and labeling the term on a diagram. The assessment also read aloud. The assessment of the science terms did not include reading comprehension or application of the science terms.
- Another limitation of this study was the lack of maintenance probes. Longer term effects of AR vocabulary instruction are needed. Time constraints prevented the collection of maintenance probes in this study.

Future Research

Future research is needed to replicate this study's methods and procedures.

- Across other disability populations and age groups.
- Investigation in other subjects areas such as reading, math, and social studies as well as functional life skill domains.
- Explore what AR features lead to the positive outcomes without distracting the learner. The following instructional AR components could be examined through a series of comparative intervention studies:
 - Length of AR content
 - using video and/or static pictures
 - the word length of definitions
 - use of audio information.
- Lastly, AR used to teach vocabulary should be compared to more established vocabulary instructional procedures, such as time delay, read alouds, and picture-to-text matching.

Summary

- The AR vocabulary intervention produced a positive impact on student mastery of the science vocabulary terms through its combination of real world and digital content.
- Using the AR vocabulary intervention was a positive experience for all the participants according to the social validity data.
- The findings of the study support further examination of AR as medium for science and vocabulary instruction for students with disabilities.

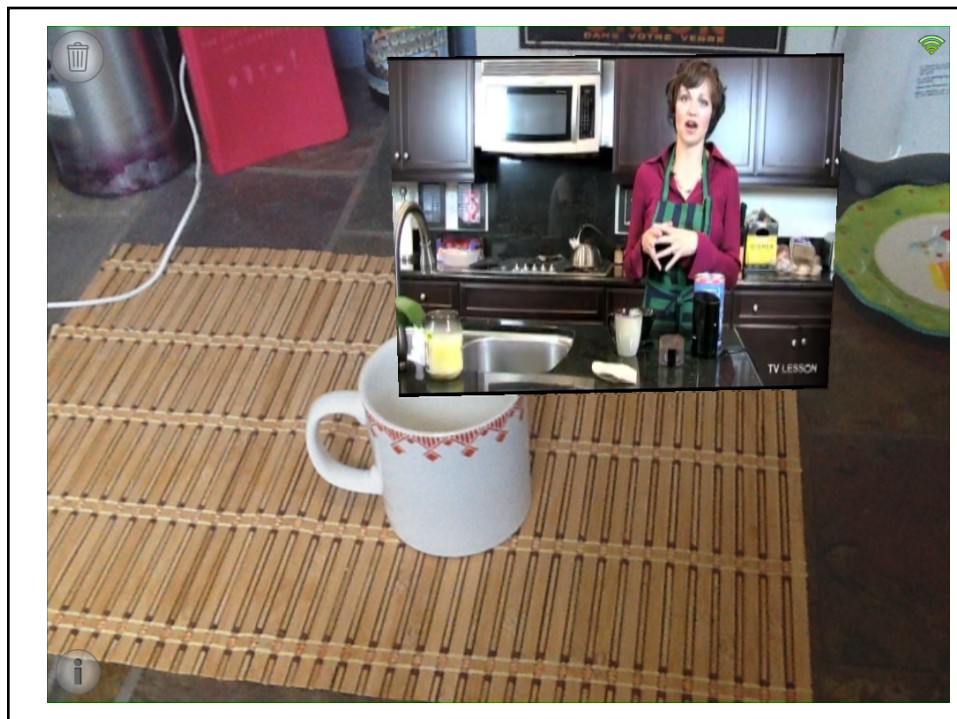
Lines of Research

Marker Based AR

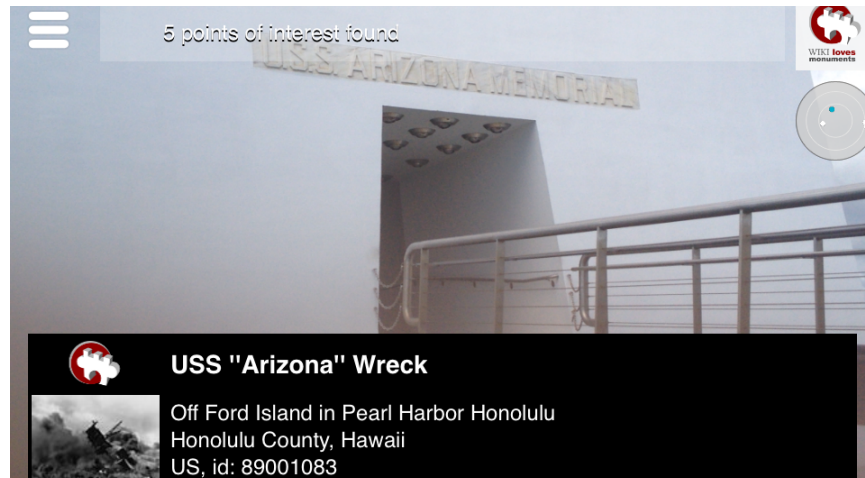
- Additional Academic Areas
- Populations
- Apply for functional skills

Markerless AR

- More functional activities
- Additional employment related activities
- Populations
- Location based academic learning



Location Based Learning



Location Based Learning



Preparing for the Augmented Future

Why does this matter?

These and other new technologies from this emerging augmented future will create additional social challenges, questions and debate. Acknowledging and addressing these challenges will become part of the domain of special education technology in the near future as AR experiences become as common a learning tool as video, interactive simulations, computers, the Internet, and mobile devices.

QUESTIONS?



THANK YOU SO MUCH FOR COMING!

Dr. Don McMahon

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@macddm on twitter

Strategy Instruction for Young Adults with
Intellectual Disability at the Postsecondary
Education Setting

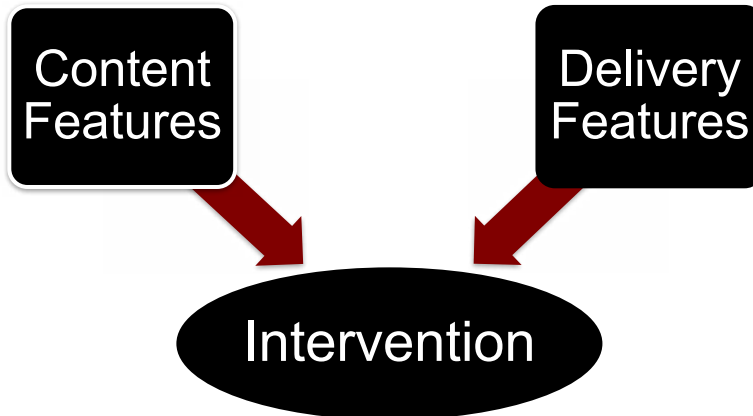
Youjia Hua, Ph.D.

Department of Curriculum & Instruction and Special Education
University of Virginia

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Department of Teaching and Learning
University of Iowa

Features of Strategy Instruction



- Content Features
 - Teach the critical steps in order to perform an academic task
 - Teach self-regulation skills
 - Questioning
 - Evaluation
 - Reinforcement
 - Coping
 - Bridge the gap between the demand of the task and skill deficit

- Delivery Features
 - Explicit instruction with sufficient opportunities to practice
 - Presentation techniques
 - Choral responding
 - Increased opportunities to respond
 - emphasized guided and independent practice
 - Used small group vs. one-on-one instruction

Strategy Development Model

Instructional Stages	Purpose
Stage 1	• Provide rationale and obtain commitment
Stage 2	• Describe the strategy
Stage 3	• Model the strategy
Stage 4	• Verbal practice and memorize
Stage 5	• Guided practice
Stage 6	• Independent practice with teacher feedback

Components of Strategy Instruction

- ☐ pretest students' skill knowledge
- ☐ description of the strategy
- ☐ modeling/practice
- ☐ self-prompting
- ☐ posttest students' skill knowledge
- ☐ generalization

Deshler and Schumaker (1986)

Study 1

What are the effects of the paraphrasing strategy on reading comprehension of expository materials for young adults with ID?

Reading Skill Deficits of Young Adults with ID

- Oral reading fluency and vocabulary knowledge
- More explicit and deliberate comprehension instruction needed

Hua, Hendrickson, et al., 2012; Hua, Therrien, et al., 2012

Paraphrasing Strategy

- 1. Read** a paragraph
- 2. Ask** myself “What was the main idea and two details?”
- 3. Put** it into my own words

Schumaker, Denton, & Deshler, 1984

Why Paraphrasing Strategy?

- Content feature
 - Reduced demand of reading comprehension on working memory
 - Prompted use of self-regulation to identify text details
 - Facilitated comprehension and memorization of reading materials

Cain, Oakhill, & Bryant, 2004; Ellis & Lenz, 2010; Hagaman, Casey, & Reid, 2010; Hagaman & Reid, 2008; Katims & Harris, 1997

Study 1 Research Question

What are the effects of the paraphrasing strategy on reading comprehension of expository materials for young adults with ID?

<i>Participants Demographic Data by Experimental Condition</i>					
Variable	Gender	Age	Ethnicity	Disability	IQ
Experimental Group					
Student One	Female	20	Asian	ID	77
Student Two	Female	20	Asian	ID	71
Student Three	Female	21	Caucasian	ID	67
Student Four	Male	21	Caucasian	ID	77
Student Five	Male	20	Caucasian	ID, ADHD	66
Comparison Group					
Student One	Female	20	Caucasian	ID	78
Student Two	Female	21	Caucasian	ID, ADHD	67
Student Three	Male	20	Caucasian	ID, ADHD	77
Student Four	Male	20	Caucasian	ID	67
Student Five	Male	23	Caucasian	ID	74

Note. IQ = full scale intelligence quotient scores reported for the Weschler Intelligence Scale for children.

Materials

- Timed Reading Series (Spargo, 1989)
 - Average reading level: 6th grade (range, 6.4-6.9)
 - Average length: 400 words (range, 399-402)
 - Each passage has five paragraphs

General Procedures

- Explicit instruction lesson plan
 - Review and overview of the lesson
 - Model
 - Prompt
 - Check
 - Close

The Paraphrasing Instruction

Instructional Stages	Purpose
Stage 1	• Provide rationale and obtain commitment
Stage 2	• Describe the strategy
Stage 3	• Model the strategy
Stage 4	• Verbal practice and memorize
Stage 5	• Guided practice
Stage 6	• Independent practice with teacher feedback

Dependent Variables

- Each passage contains 5 paragraphs
 - 5 main ideas
 - 10 details
- Scoring criteria:
 - Must contain a complete sentence;
 - Must be accurate;
 - Must be relevant to the paragraph
 - Must contain new information;
 - Must be meaningful;
 - Must be paraphrased.

For thousands of years, the camel has helped people live in the deserts of Asia and Africa. It can travel great distances over hot sands for days without water. It can carry a person or a load of freight. For this reason, it is sometimes called the "ship of the desert." **STOP**

The camel supplies food and many valuable materials to desert dwellers. These people can live for many weeks on thick, cheesy camel's milk and on the meat of young camels. Desert dwellers make camel's hair into tents, blankets, rugs, clothing, and rope and more. Dried camel droppings supply fuel for cooking fires. When a camel dies, its hide is used for making sandals, water bags, and many other necessities. **STOP**

There are two kinds of camels. The dromedary, or Arabian, camel has a single hump on its back. The Asian camel has two humps. The dromedary once roamed wild but is now found only in domestication. Groups of them, however, are often left on their own for up to five months. The Asian camel is primarily a domesticated animal, but small wild herds are still found in areas of Mongolia and China. Only about 300 to 500 Asian camels still live in a wild state, so the species is considered in danger of extinction. **STOP**

Although often ill tempered, the camel is wonderfully adapted for the work it has to do. No other animal can live and work on such scant supplies of food and water in so hot and dry a climate. One of the few things that a camel will do on command is kneel. It is easier for a person to climb onto or load up the animal when it kneels. The camel seldom works without a protest. The uproar in a camel yard when a caravan is being loaded is deafening. **STOP**

The camel's most striking feature is the large hump or humps on its back. The hump is formed of fat and muscle. When a camel is well fed and given enough water, the hump is erect. If the camel has to go without food and water for a period of time, the hump becomes limp and leans to one side. The camel's body is covered with a shaggy, sand-colored coat. The hair sheds in great handfuls, giving the animal a perpetually frowsy look. Long eyelashes protect the eyes from sandstorms and the glare of the desert sun.

Procedural Integrity and Reliability

- Procedural integrity = 100%
- Main idea reliability = 100%
- Detail reliability = 97%

Results

Total Number of Main Ideas and Details Recalled by Participants in the Control and Experimental Group

Variable	Control		Experimental	
	Pretest	Posttest	Pretest	Posttest
	<i>M</i> (SD)	<i>M</i> (SD)	<i>M</i> (SD)	<i>M</i> (SD)
Main Idea	.80 (.84)	.40 (.55)	1.00 (1.00)	3.00 (.71)
Details	.40 (.89)	.80 (.84)	.60 (1.34)	4.40 (1.67)

STUDY 2

The EDIT Strategy

(Hughes, Schumaker, McNaughton, Deshler, & Nolan, 2010)

Woods-Groves, S., Hua, Y., Ford, J. W. & Neil, K. M. (2017). Efficacy of an electronic editing strategy with college students with intellectual and developmental disabilities. *Education and Training in Autism and Developmental Disabilities* , 54(4), 422-436.

Why Teach Editing Skills?

The writing process has been described by some as a cyclical process:

- ❖ planning
- ❖ production of text
- ❖ revising

(Hayes & Flower, 1987; Hayes, 1996)

Revising Process: Editing Skills

- Students who have difficulties in written expression. Spend time excessive time.....
- Creating content
- Correcting grammatical errors
- Impedes further text development
- revising beyond grammatical errors
- *Common difficulties in revising/editing skills:*
 - spelling errors
 - capitalization & punctuation errors
 - overall appearance
 - substance

(Archer & Hughes, 2010; Hughes et al., 2010; MacArthur et al., 1995; McNaughton et al., 1997)

☐ Enter Your First Draft ☐ Do a SPELL Check. Put cursor at beginning ☐ Read the sentence with the error out loud ☐ Select the correct option * If you don't see the correct option ☐ 1. Say the word aloud. ☐ 2. Add any letters that you hear. ☐ 3. Run the spellchecker on the word. ☐ 4. If you see the correct option, select it. ☐ Pass over the document ☐ Express each sentence ☐ Look for homophone errors ☐ Look for typos	☐ Interrogate yourself Use the COPS Questions ☐ Put the cursor at the end of the document ☐ Capitalization questions ☐ Look at the beginning of sentences ☐ Proper Nouns (people, places, and things) ☐ Overall appearance questions ☐ look at paragraph indention ☐ Spacing between sentences ☐ Punctuation questions ☐ Is there punctuation at the end of the sentence? ☐ Is it correct? ☐ Substance questions ☐ Read the sentence out loud. ☐ Is the sentence missing any words. ☐ Does it make sense? ☐ Type in corrections and run spellchecker
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(Hughes, Schumaker, McNaughton, Deshler, & Nolan, 2010)

EDIT Research Questions

1. Will there be a significant difference in the total number of editing errors corrected and in the type of errors corrected for students who received the EDIT instruction compared to students who did not receive the instruction?
2. Will students who were taught the strategy maintain their skills over time?

Participants *N* = 15

Educational Services:

Autism	1 (7%)
PDD	3 (20%)
ID	6 (40%)
Multiple Disabilities	4 (26%)
Down Syndrome & ID	1 (7%)
Gender: 6 (40%) Females; 9 (60%) Males	

Participants

➤ Demographic

Across 8 states (rural, suburban, urban)

Race = White

Ages: 18 to 23 years (*M* = 19.40, *SD* = 1.40)

WJIII Total Score 20 – 90, (*Mdn* = 74)

WJIII Broad Reading 30 – 87, (*Mdn* = 78)

WJIII Broad Writing 30 – 95, (*Mdn* = 79)

Experimental Sequence

1. Students randomly assigned to the intervention (7) or control group (8).
2. Administered counter-balanced EDIT pretests.
3. Students in treatment group (11 sessions, 45 min sessions in group instruction) twice a week (total of 8.25 hrs.)
4. The posttest measure was administered to the treatment/control group 1 wk following end of intervention.
5. An EDIT Scoring Key was used to evaluate pre/post tests.
6. Maintenance probe

Pre/Post Prompts

- Were from the EDIT manual but were adapted for 3rd grade reading level.
- **2 prompts:**
 - ☐ **Giant Panda**
 - ☐ **Redwood Trees**
- 240 word avg.
- 3 paragraphs

Pre/Post Prompts

➤ 30 Errors

- ☐ 6 Spelling
- ☐ 6 Capitalization
- ☐ 6 Overall Appearance
- ☐ 6 Punctuation
- ☐ 6 Substance

Material

- Instructor's manual
- Computers, memory sticks
- Student folders
 - progress graph
 - daily lesson materials
 - graphic organizers
- Highlighters
- **EDIT** mnemonic/front folders

Daily Lesson Structure

➤ Included:

- explicit instruction
- modeling
- thinking out loud procedures
- immediate feedback
- daily probes for mastery

L	# Sa	# Sb	Instruction Overview
	16	11	Topic/ Content
1st	1	1	• Rational for editing one's work * Introduced EDIT Strategy
2nd	7	4	* Taught "Examine your 1 st draft" & "Do Spell Check"
3rd	3	2	* Taught "Interrogate yourself with COPS" * Taught "Type in corrections and run spellchecker" * Practiced steps E, D, I, and T together.
4th	1	1	* Rapid fire practice and internalize the mnemonic & Practice EDIT steps
5th	4	3	* Practiced using EDIT steps (guided & independent practice)
For the 2 nd - 4 th lessons the following sequence was employed: (Demonstrate / Guided Practice with feedback/ Independent Practice)			

Dependent Measures

Pre/post test essays

- Giant Panda
- Redwood
- Evaluated using **EDIT Scoring Keys**.
 - ❖ Overall score total correct
 - ❖ 5 separate (i.e., spell, cap, overall, punc, & substance) # errors corrected totals

EDIT Posttest Scores

Posttest	EDIT Total	Spelling	Capitalization	Overall App	Punctuation	Substance
C group (8)	13.75* (8.27)	4.00* (2.51)	3.81* (2.14)	3.63* (2.71)	2.13* (1.64)	0.31* (0.46)
T group (7)	20.21* (7.15)	5.36* (0.99)	4.57* (1.13)	4.14* (2.27)	4.07* (2.35)	2.14* (1.49)
ES	0.84	0.71	0.44	0.20	0.96	1.66
ANCOVA	$F(1, 15) = 11.049$ $p = .006^{**}$	$F(1, 15) = 6.854$ $p = .022^{**}$	$F(1, 15) = .674$ $p = .428^{ns}$	$F(1, 15) = .495$ $p = .495^{ns}$	$F(1, 15) = 6.099$ $p = .030^{**}$	$F(1, 15) = 10.708$ $p = .007^{**}$

EDIT Strategy

☐ **Students in intervention group corrected more errors: (Significant)**

- Total ($d = .84$)
- Spelling ($d = .71$)
- Capitalization *Not Sig* ($d = .44$)
- Overall App. *Not Sig* ($d = .20$)
- Punctuation ($d = .96$)
- Substance ($d = 1.66$)

EDIT 5 Weeks Later

- **Maintenance** 5 wks/ end of intervention
- $n = 15$ (T = 7, C = 8)
- No review of EDIT Strategy

- | | |
|---|---|
| <ul style="list-style-type: none"> ➤ <u>Significant</u> ➤ Total Errors Corrected ➤ Spelling ➤ Punctuation ➤ Substance | <ul style="list-style-type: none"> <u>Not Sig</u> Capitalization Overall App. |
|---|---|

EDIT 12 Weeks Later

- **Maintenance** 12 wks/ end of intervention
- No review of EDIT Strategy

- | | |
|--|--|
| <ul style="list-style-type: none"> ➤ <u>Significant</u> ➤ Total Errors Corrected ➤ Spelling ➤ Punctuation | <ul style="list-style-type: none"> <u>Not Sig</u> Capitalization Overall App. Substance |
|--|--|

Discussion

- Two studies
- Effect of Strategy Instruction
- Utility of Strategy Instruction across content
- Strategy Instruction in PSE

Questions



The TPSID National Coordinating Center at Think College is a project of the Institute for Community Inclusion at UMass Boston.
The Center is funded by the Office of Postsecondary Education Grant #P407B100002.



Have you joined our Affinity Group on Research?

- Learn about research that is happening in the field of inclusive higher education
- Connect with other researchers and explore possible collaborations
- Share tools and resources

Next meeting: Wednesday January 31st 2:00-3:00pm Eastern

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Have you joined the Conversation?

Think College has opened a new forum to encourage the exchange of ideas, problem-solving, and resource sharing between individuals interested in inclusive post-secondary education. Using the Slack platform, we are able to categorize conversations by topic, share files and other information, and talk in groups or with individuals.



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And once you're logged in, be sure to join the #research channel!



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Thank you for joining us today!

- Please make sure to complete the evaluation
- Join us on Slack to continue the conversation in the #research channel
- Interested in presenting at a future Research Summit?
 - Send an email to clare.papay@umb.edu



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